AirInk

A Python finger-tracking whiteboard

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What is AirInk?

A webcam becomes a pen.

Raise your index finger to write.

Raise two fingers to move.



Existing approach & AirInk

Mouse / touch drawing

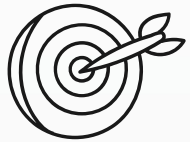
Pointer controls the pen.

Precise, familiar input.

AirInk

Finger pose controls the pen.

Touch-free camera input.



A few useful controls



Ink tools

Pen, eraser and six colours



Brush size

Thin notes or bold strokes



Two views

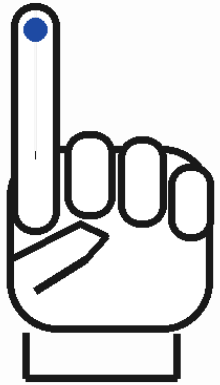
Zoom, opacity or whiteboard



Save & reopen

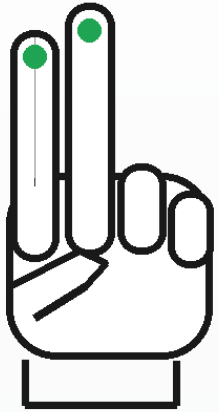
Keep and continue drawings

One finger writes. More fingers pause.



DRAW / ERASE

Only index raised



MOVE / PAUSE

Two or more raised

How it works

1

Camera

Track the full camera frame

2

Hand

Find 21 landmarks

3

Pose

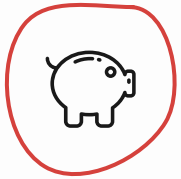
Check which fingers are up

4

Draw

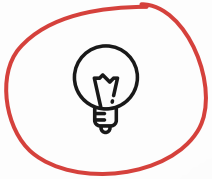
Join fingertip positions

The code to remember



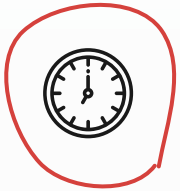
writing_pose()

Check five fingers



tick()

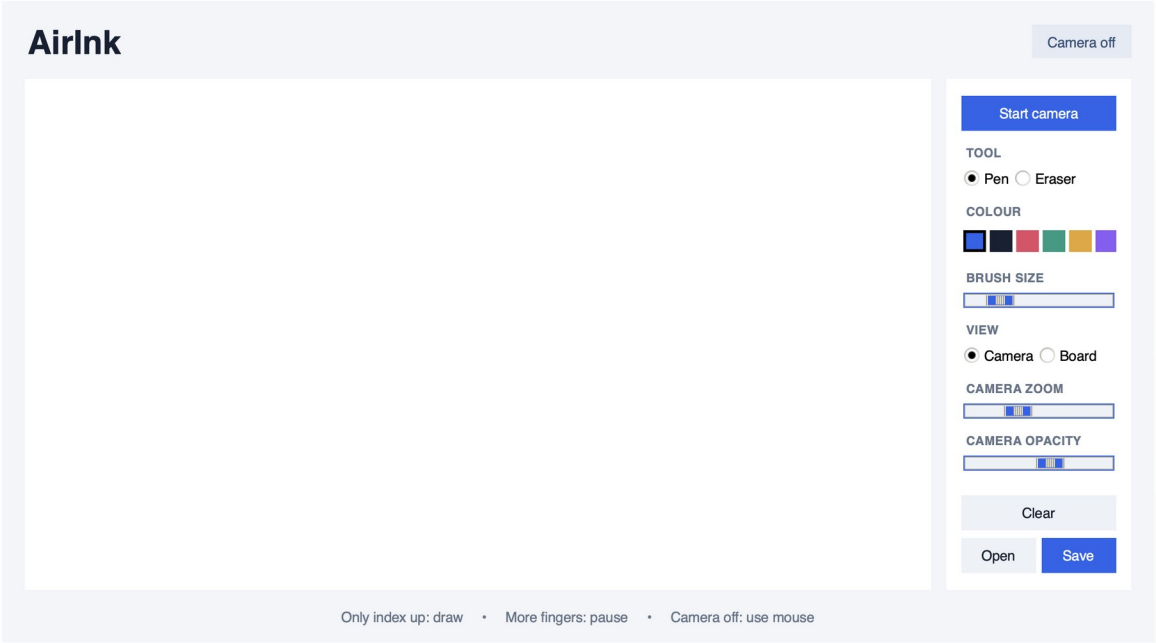
Track and refresh



draw()

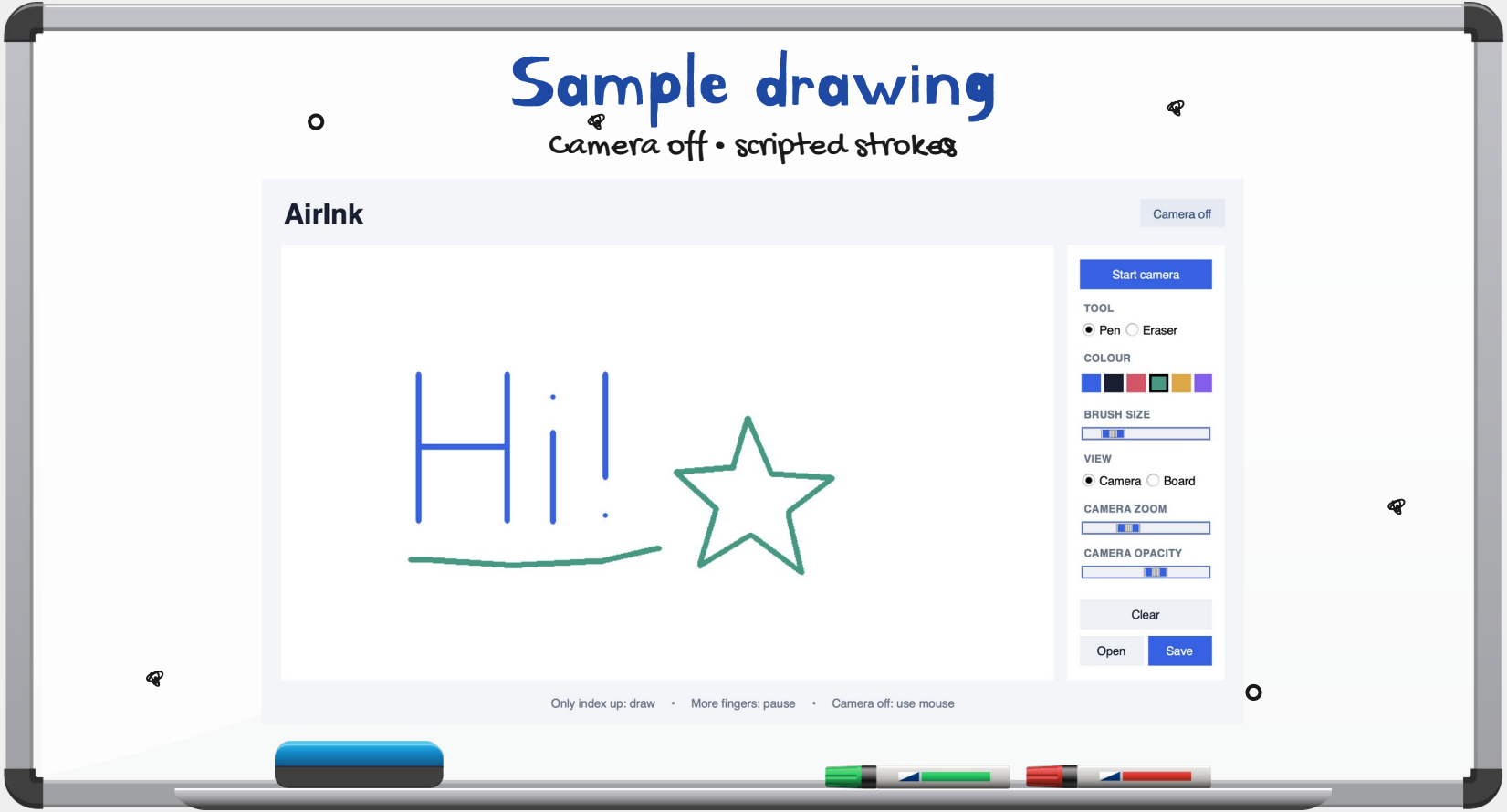
Connect two points

The application



Sample drawing

Camera off • scripted strokes



Testing & next steps

38 tests passed

Pose, ink, PNG and error handling

✓

21

Model checked

21 landmarks on a sample hand

Try in class

Check light, camera and finger pose

!

+

Improve later

Undo and rotation-tolerant gestures

Thank you!

Questions?

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Presentation template: Slidesgo

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